

Mustafa Talha Avcu

New York, NY
itsavcu@gmail.com
(323) 369-1727

WORK EXPERIENCE

- **Amazon** Seattle, WA - Toronto, ON
Software Development Engineer II August 2022 - June 2026
 - Implemented the guardrail layer within a new LLM orchestration service built to integrate generative AI capabilities into Amazon's customer service chatbot, validating customer inputs and model outputs to protect against adversarial inputs, inappropriate responses, and unsupported claims. Contributed to the launch of Amazon's first LLM-powered customer support experience, initially supporting Amazon Music customer issues.
 - Designed and built reusable ML infrastructure using AWS CodePipeline, S3, and SageMaker, enabling automated training, testing, and deployment of ML models through CI/CD pipelines. Used the infrastructure to deploy a model that infers customers' issues from historical data and routes them to specific support, reducing average chatbot interaction time by 9%(440s to 400s) with no negative impact on other metrics.
 - Led scaling efforts for the Retailer Ad Service (RAS) catalog and ingestion services during Macy's onboarding to support catalogs with 2M+ offers uploaded every 2 hours. Built cross-service dashboards to diagnose throttling and exceptions, implemented slicing and enforced a limit on concurrent ingestion batches, and modified retry configs, resolving ingestion issues and enabling stable onboarding of RAS's largest retailer to date.
 - Owned and implemented the RAS core services integration for Amazon Demand-Side Platform (ADSP), extending Retailer Ad Service (RAS) attribution to support both Sponsored Products (SP) and ADSP. Built ingestion jobs and unified tables to normalize campaign and traffic data from separate SP and ADSP upstream sources, and independently developed the unified attribution query to perform attribution across both programs.
- **Neural Systems Engineering and Information Processing Lab (NSEIP)** Los Angeles, USA
Research Assistant Sept 2019 - Dec 2021
 - Modelled large-scale brain networks during direct electrical stimulation, and developed nonlinear models that are amenable for control in closed-loop brain computer interface (BCI) applications.
- **Agency for Science, Technology and Research (A*STAR)** Singapore
Data Science Intern Jun 2018 - Sept 2018
 - Developed **SeizNet**, a CNN-based seizure detection model, achieving 95.8% sensitivity with 0.17 false alarms/hour on EEG data from 29 patients; demonstrated that a 2-channel configuration achieved 93.3% sensitivity, outperforming a traditional SVM baseline using the full set of EEG channels.

TECHNICAL SKILLS

Programming Languages: Java, Python, Scala, JavaScript, TypeScript, SQL, C++

Machine Learning and Data: Spark, Large Language Models (LLMs), Keras, TensorFlow, Scikit-learn

AWS Technologies: AWS CDK, CloudFormation, CodePipeline, S3, Lambda, ECS, SQS/SNS, CloudWatch, Clean Rooms, SageMaker, Bedrock

Others: Linux, Docker

EDUCATION

- **University of Southern California** Los Angeles, CA
Master of Science in Computer Science, CGPA: 3.96/4.00 May 2022
 - Selected Courses: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing
- **Bilkent University** Ankara, Turkey
Bachelor of Science in Electrical and Electronics Engineering, CGPA: 3.93/4.00 Jun 2019
 - Exchange student at Korea University for one semester

HONORS & AWARDS

- **Two-time Amazon org-wide hackathon winner among 20+ competing teams**
- **Annenberg Fellowship with admission to University of Southern California (USC) PhD program, 2019**
- **Academic Excellence Graduation Award, 2019** Bilkent University Electrical and Electronics Engineering Department
- **IEEEExtreme Programming Competition**
 - **IEEEExtreme 10.0, 2016:** Ranked 2nd in Turkey, 20th in IEEE Region 8, and 65th overall
 - **IEEEExtreme 9.0, 2015:** Ranked 3rd in Turkey, 30th in IEEE Region 8, and 82nd overall
- **Nationwide University Entrance Exam (LYS):** Ranked in top 0.01% among 2 million students in Turkey, 2014

PUBLICATIONS

M. T. Avcu, Z. Zhang, and DWS. Chan "Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks" IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019